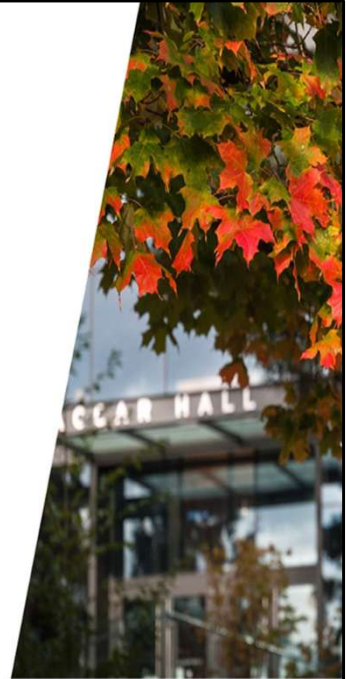


ACCOUNTING & FINANCE FUNCTION 2

Data Analytics for Professional Accountants
(ACCTG 522)

Class 14 | MPAcc Class of 2026

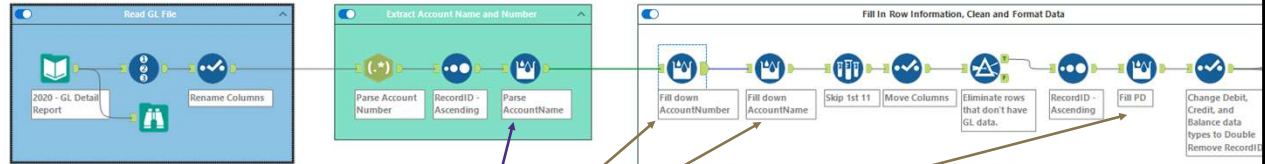


Welcome!

Class Agenda

- **Review: Multi-Row Formula tool**
- Lab:
 - Case 2. Vendor Analysis / Suspect Vendors
 - Case 3. PO Splitting
- Conclusion and look ahead

Multi-Row Formula Tool



1. Parse related field like row1:number, row2:name
2. Fill in missing values within a field (i.e., column)

Welcome!

Class Agenda

- Review: Multi-Row Formula tool
- **Lab:**
 - **Case 2. Vendor Analysis / Suspect Vendors**
 - Case 3. PO Splitting
- Conclusion and look ahead

Overview

The U.S. Department of the Treasury's Office of Foreign Assets Control (OFAC) oversees U.S. economic and trade sanctions. These sanctions target certain foreign countries, known terrorists, narcotics traffickers, and other threats. OFAC strongly encourages companies to have a sanction compliance program that includes internal controls and auditing.

To ensure compliance with OFAC policies, invoices authorized for payment are checked against the OFAC Consolidated Screening List (CSL), which is a list of sanctioned individuals or entities maintained by the Office of Foreign Assets Control. Identify whether any of the provided vendors are on the OFAC CSL.

Objective

Determine whether any of the vendors are on the OFAC CSL. Create a single output file with all of the vendors that are on the list separated into three tabs based on the vendor's status.

Additional note: In a real-world process we would want to check for names that are similar but slightly different (think Fred vs. Freddy). In this example, exact matches are all we are looking for, but in some instances companies might want to utilize fuzzy matching to identify pairs that are close but not exact.

Processes

1. Pull in the Vendors.yxdb file, clean the names by removing punctuation and standardizing the case, and ensure there is only one record for each Vendor ID.
2. Pull in the Consolidated Screening List – Modified.yxdb, clean the names by removing punctuation and standardizing case, and add a unique identifier to each record.
3. Identify the vendors that are present on the OFAC list, and organize those vendors by date with the newest vendors first. Remove any fields that are entirely blank.
4. Output an Excel file of those vendors with a separate tab for each status.

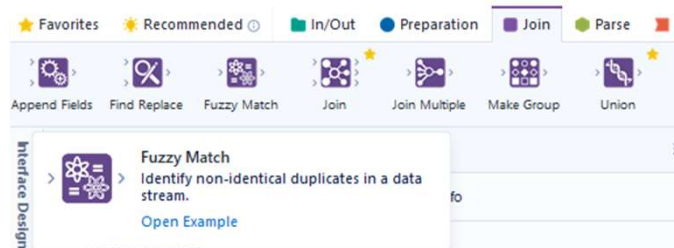
Starter files

 Consolidated Screening List - Modified.yxdb	9/26/2023 4:29 PM	Alteryx Database	1,577 KB
 Suspect Vendors - No Hints.yxml	11/10/2025 8:19 AM	Alteryx Workflow	65 KB
 Suspect Vendors - Tool Hints.yxml	8/18/2022 7:51 AM	Alteryx Workflow	83 KB
 Vendors.yxdb	9/26/2023 4:29 PM	Alteryx Database	399 KB

Extension

Determine whether any of the vendors are on the OFAC CSL. Create a single output file with all of the vendors that are on the list separated into three tabs based on the vendor's status.

Also attempt: *In a real-world process we would want to check for names that are similar but slightly different (think Fred vs. Freddy). In this example, exact matches are all we are looking for, but in some instances companies might want to **utilize fuzzy matching to identify pairs that are close but not exact.***



Are any of these names

Data Dictionary

Vendors.vyxdb

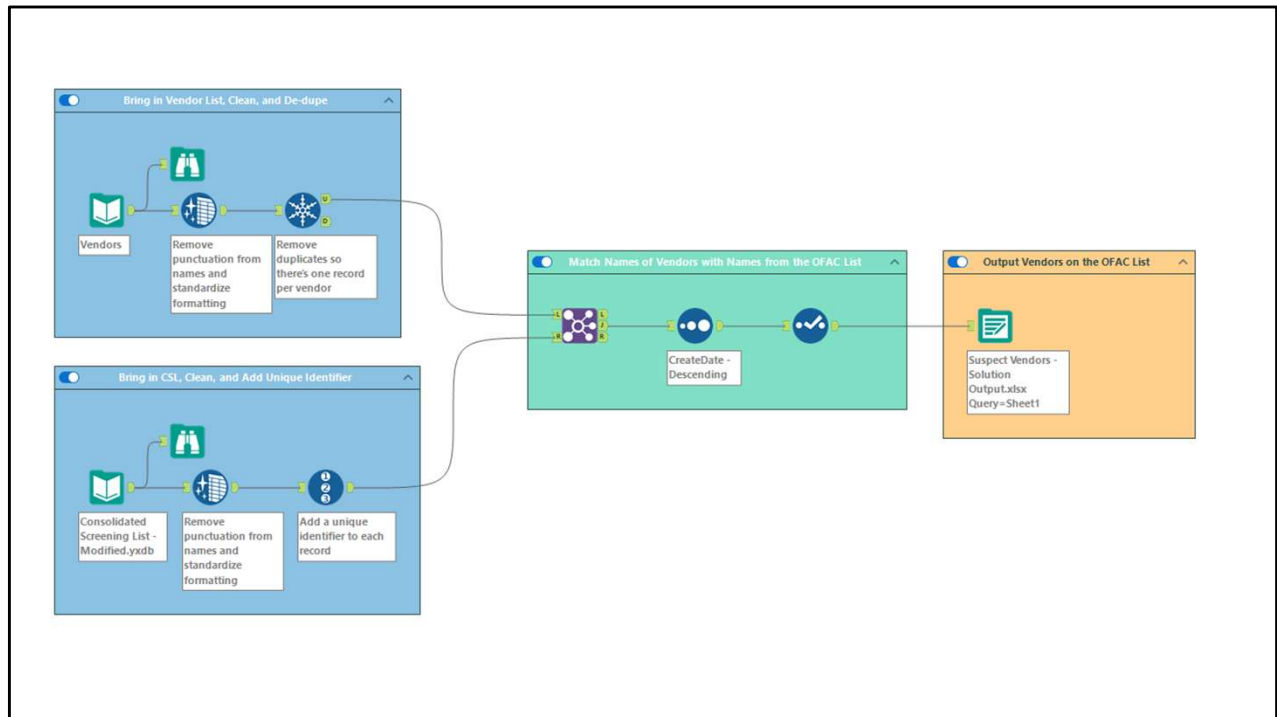
Column Name	Data Type	Description
<u>VendorID</u>	V_String	
<u>CreateDate</u>	DateTime	
<u>Status</u>	V_String	
<u>Vendor Name</u>	V_String	

On this list?

Consolidated Screening List – Modified (CSL)

Government list modified to remove some extraneous columns

Column Name	Data Type	Description
<u>source</u>	V_WString	
<u>entity_number</u>	V_WString	
<u>type</u>	V_WString	
<u>programs</u>	V_WString	
<u>name</u>	V_WString	Name you are matching to
<u>title</u>	V_WString	
<u>addresses</u>	V_WString	
<u>federal_register_notice</u>	V_WString	
<u>start_date</u>	V_WString	
<u>end_date</u>	V_WString	
<u>license_requirement</u>	V_WString	
<u>license_policy</u>	V_WString	
<u>remarks</u>	V_WString	
<u>alt_names</u>	V_WString	
<u>citizenships</u>	V_WString	
<u>dates_of_birth</u>	V_WString	
<u>nationalities</u>	V_WString	
<u>places_of_birth</u>	V_WString	



Welcome!

Class Agenda

- Review: Multi-Row Formula tool
- **Lab:**
 - Case 2. Vendor Analysis / Suspect Vendors
 - **Case 3. PO Splitting**
- Conclusion and look ahead

Overview

In most companies, employees with purchasing authority have a monetary limit on how much they can purchase without management authorization. For example, an IT manager may be able to purchase up to \$50,000 in software and equipment. Any amount over that would require director approval. In practice, this means that the IT manager requires director approval for any Purchase Order (PO) with a purchase amount over \$50,000. If a manager wants to purchase \$75,000, they could try to get around this by issuing two POs to the same vendor totaling \$75,000 but each being less than \$50,000. This is a violation of corporate compliance rules known as "PO Splitting."

Objective

Create a simple workflow to detect PO splitting. In this case we are looking for multiple POs to the same vendor within a three-day period. At this point we are not checking to see if the same buyer is issuing the POs. (It is possible that there is collusion between employees to hide the splitting so there would need to be further examination of the POs found by our analysis).

Process

1. The input file has PO line-item detail. The first step is to remove inactive PO's, eliminate fields specific to the PO line, and retain only one record for each PO.
2. Then identify the vendors to whom multiple POs have been issued.
3. Determine if there are any vendors where the interval between their consecutive transaction dates is 3 days or less.
4. Output this list for further investigation.

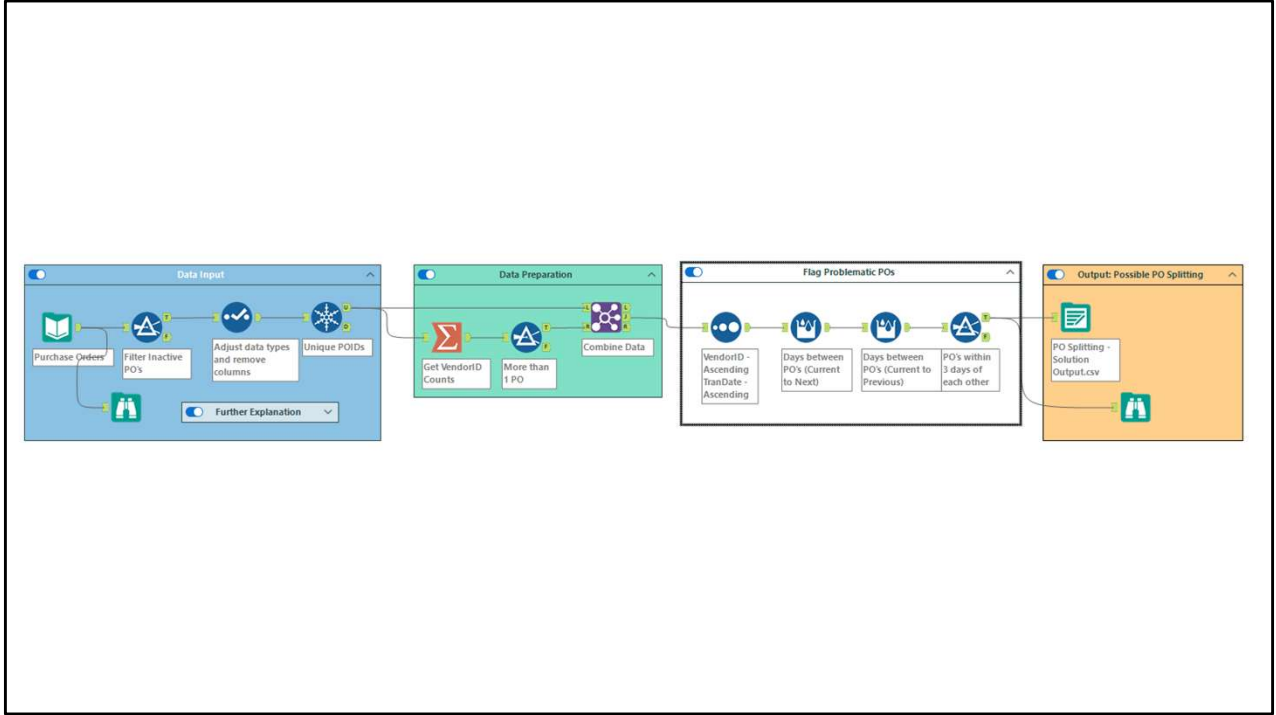
Starter Files

 PO Splitting - No Hints.yxmd	11/10/2025 8:24 AM	Alteryx Workflow	74 KB
 PO Splitting - Tool Hints.yxmd	11/10/2025 8:21 AM	Alteryx Workflow	89 KB
 PO Splitting-Workflow Started.yxmd	11/10/2025 8:31 AM	Alteryx Workflow	88 KB
 Purchase Orders.xlsx	9/26/2023 4:29 PM	Microsoft Excel W...	6,843 KB

Data Dictionary

Purchase Orders.xlsx

Column Name	Data Type	Description
POID	Double	Purchase Order ID
BuyerName	V_String	PO originator (buyer)
CloseDate	Date	PO close date
CompanyID	V_String	Company division ID
CreateDate	DateTime	Date PO created
InvoicedAmt	Double	Total cash amount invoiced to data
OrderedQuantity	Double	Quantity ordered (line item)
IsClosedForInvoiced	V_String	PO Status
IsClosedForReceiving	V_String	PO Status
PurchAmt	Double	Total cash amount of invoice
TranStatus	V_String	Transaction status (Open if field is empty)
SalestaxAmt	Double	Sales tax on entire PO
TranAmt	Double	Transaction amount
TranDate	Date	Transaction date
TranNo	V_String	Transaction number
TranType	V_String	= "Purchase Order (Standard)"
VendorID	Double	Vendor ID
PaymentTerm	V_String	Payment terms
DueDayOrMonth	Double	Due date
PmtTermsID	V_String	Payment terms code
VouchID	Double	Voucher ID (line item)
POLineKey	Double	Identifier for PO Line (line item)
POLineNo	Double	PO line number (line item)
IsReceivingRequested	Double	How to receive item (line item)
POLineStatus	V_String	PO Line Status (line item, but needed for initial cleansing-- if any line is active, the PO is active)
TargetCompanyID	V_String	Varies by line item
UnitCost	Double	Unit cost (line item)
UnitMeasure	V_String	Unit measure (line item)



Conclusion and look-ahead

Next 2 classes:

- Accounting and finance function analytics part 3 using Alteryx (1 class / 1-2 longer cases).
- Alteryx Escape Room Challenge.

Thank you!

